Sushanth Paatnaik

Inventor · Deep-Tech Founder · Materials Strategist

Six-time Indian Presidential awardee building graphene, nano-materials, and industrial deep-tech ventures from India.

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Profile

Inventor, deep-tech founder, and six-time Indian Presidential awardee.

An inventor quietly building industrial futures. Born in Bhubaneswar, Odisha, the work began in a borrowed workshop at fourteen — and has not really paused since. An industrial systems builder, deep-tech founder, and materials strategist, building from India for the world.

Fifteen years inside the workshop have taught one lesson without exception: frontier science only matters when it reaches the industrial world. The brief is not a paper, not a prototype, not even a product — it is the discipline of carrying invention through capital, through manufacturing, and through the long quiet years before scale.

The philosophy is simple: Engineer matter. Engineer capital. Engineer scale. Hold all three in one hand long enough for the work to outlive the inventor. The direction is narrower still — graphene and advanced materials, patient industrial capital, and a vertically integrated stack designed to make Indian deep-tech globally inevitable in the carbon century ahead.

At a Glance

27 HONORS OF RECORD	**06** PRESIDENTIAL AWARDS	**60+** KEYNOTES SINCE 2010
23 GRAPHENE INNOVATIONS	**06** OPERATING VENTURES	**14+** YEARS OF RESEARCH

Education

BSc. — IISER Bhopal

Admitted under the KVPY-SP scholarship. Foundation in chemistry, physics, and the discipline of asking questions matter cannot easily answer.

BEd. (ETE) — OCT, Bhopal

Special Achiever category — a conviction that an inventor who cannot teach has not really finished the invention.

Ventures

Founder, co-founder and chief innovation officer across six operating vehicles — plus an advisory roster of five houses across industry, materials and climate. A closed loop from atom to invoice.

Monoatom Labs ***— Co-Founder & CEO · 2025 · Materials***

A scalable and economical method to manufacture graphene — and the applications that turn it into real-world performance gains. (monoatomlabs.com)

Grafillium ***— Co-Founder & CIO · 2025 · Eco nano additives***

Deep-tech nanomaterial additive technologies that boost efficiency and cut carbon emissions across power, logistics and heavy industry. (grafillium.com)

SPI Industries ***— Founder & CEO · 2024 · Industrial Systems***

R&D-led industrial solutions in nanomaterial engineering — translating advanced materials science into deployable systems. (spiindustries.co)

InThinks *— Co-Founder · 2026 · Innovation Studio*

An ideation and innovation studio that shapes early-stage thinking into products, then transfers or licenses the technology to partner organisations. (inthinks.com)

Starunico Capital *— Co-Founder · 2026 · Deep-Tech Capital*

Backing founders building the materials and energy layer of the next century. (starunico.com)

Magppie *— Chief Innovation Officer · 2025 · Design + Living*

Pioneering the world's first 100% stone-built modular kitchen — transforming ordinary homes into wellness homes. (magppie.com)

Advisory Roster

- Vinrox — Materials
- VPRPL — Industrial Systems
- WeHear — Consumer Tech
- Tileopedia — Surface Technologies
- Magppie — Design + Living

Innovations — 23 Graphene Products

One material platform, twenty-three industrial expressions — from bench formulation through plant pilot to field deployment. Graphene: ~200× the tensile strength of steel, ~13× the thermal conductivity of copper, 2,630 m²/g surface area.

Graphacrete — graphene admixture for high-performance concrete.

Graffisol — self-cleaning solar coating, +10–12% annual yield.

Aquamax — world-first system recovering 95%+ of cooling tower plume water.

Commercial · Field-Deployed

- **Graphacrete** — Graphene admixture for high-performance concrete. 49.5 MPa, −40 kg/m³ cement. (Patent)
- **Graffisol** — Solar coating: +10–12% annual yield, panel cooling, self-cleaning. (Patent)
- **Ceraphene** — Graphene ceramic coating, 9H+ hardness at one-third the premium price. (Patent)
- **HD-G-PE** — Graphene masterbatch: +30% tensile, 100× barrier. (Patent)
- **Graphenodes** — Graphene polymer cathode/anode for high-density batteries.
- **Ignitron D** — Diesel combustion optimization: 25% efficiency, 20% lower emissions.

Pilot · Plant & Field

- **Coalorix** — Coal combustion optimization: 15% utilization, 35% lower emissions.
- **Aquamax** — World-first system recovering 95%+ of cooling tower plume water.
- **Ignitron P** — Petrol combustion: 15% efficiency improvement.
- **Lubritron** — Engine oil additive: up to 6% fuel savings, 40% wear reduction.
- **Rustene** — Graphene anti-corrosion shield for steel, marine and industrial assets.
- **Gryogen** — Graphene hydrogen-selection membrane for clean fuel.
- **Mariphene** — Low-energy graphene desalination membrane.

- **Aerophenter —** Atmospheric water harvesting via graphene surfaces.
- **Fibrasphene —** Reinforced graphene glass fibres for stronger composites.
- **Voltaphene** — Graphene-enabled grid and mobility energy storage.

R&D · Bench

- **Armophene —** Graphene ballistics: lighter, stronger personal/vehicle armour.
- **Hydrocell** — Graphene hydrogen fuel cell stack for clean power.
- **Bitumax —** Bitumen additive: 1.5–2× pavement life.
- **Pyronex —** Multi-functional paint additive: fire, heat, UV, microbe shield.
- **Graphyre —** Graphene performance tyres: longer life, better grip.
- **Graphosite —** Ultra-light, ultra-strong structural graphene composites.
- **Thermaphene —** Smart thermal fabrics with active temperature regulation.

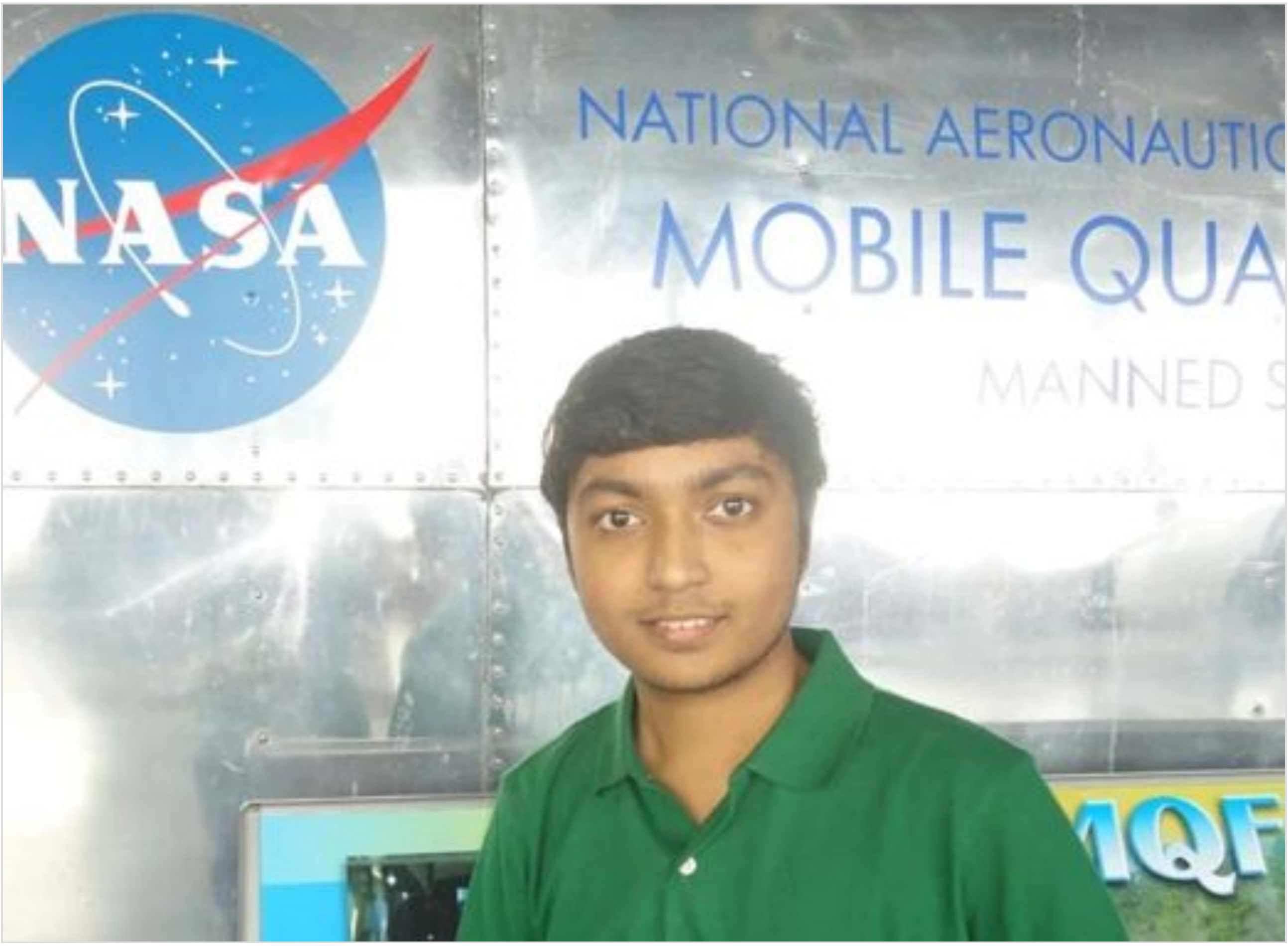
Recognitions

From a school-bench prototype to Rashtrapati Bhavan and Silicon Valley — twenty-seven honors of record and sixty-plus keynotes.

Six-time Indian Presidential awardee — recognized by three Presidents of India (2008–2013).

With Dr. A.P.J. Abdul Kalam — Presidential Award, National Innovation Foundation.

TED-India Speaker (2012) — among the youngest ever featured.



NASA recognition — Kennedy Space Center / U.S. Space & Rocket Center, Huntsville (2011).

Startup20 at G20 India Presidency, Gurugram (2023).

Defining Honors

- **Six-Time Indian Presidential Awardee (2008–2013) —** Recognized by three Presidents of India: Dr. A.P.J. Abdul Kalam, Smt. Pratibha Patil, and Shri Pranab Mukherjee, via the National Innovation Foundation.
- **MIT TR-35 Awardee (2010) —** Among the world's leading young innovators under 35 (MIT Technology Review).
- **Intel IRIS National Recognition (2010) —** Best Popular Invention for the breath-operated wheelchair.
- **NASA Recognition (2011) —** International recognition at Kennedy Space Center for human–machine interface engineering.
- **TED-India Speaker (2012) —** One of the youngest speakers ever featured.
- **MIT Fab-10 & Fab-11 Awardee (2013–14) —** MIT Center for Bits and Atoms.

Selected Achievement Ledger

- 2008 — Presidential Award (Dr. A.P.J. Abdul Kalam, NIF); Jawaharlal Nehru & CBSE National Science Exhibition Awards.
- 2009 — Presidential Awards (Pratibha Patil & A.P.J. Abdul Kalam, NIF); KVPY Fellow; NCSC Award.
- 2010 — MIT TR-35; INK Fellow; Intel IRIS Best Popular Invention.
- 2011 — NASA Award; DLF–Pramerica Spirit of Community Award; Eureka-11 IIT-Bombay.
- 2012 — Golden Book of World Record; TED-India Speaker; IDEAS-12 IIT-Kanpur.
- 2013 — Presidential Award (Pranab Mukherjee, NIF); ICAI Abu Dhabi Speaker.
- 2014 — Under-35 CEO Award.
- 2021 — STPI-Chunauti Winner.
- 2024 — ELECRAMA Winner (IEEMA); Silicon Valley Speaker.
- 2025 — CEO Club Speaker, Hyderabad.

Diplomatic & institutional engagement includes BRICS roundtables, Startup20 at G20 India, the Ministry of Power / Bureau of Energy Efficiency, DRIIV (PSA, Govt of India), and the embassies of Italy and Mauritius.

Mission

Built in India. Designed for the world. India has the talent, the demand, and the urgency to lead the carbon century. What remains is patience — the institutional patience to translate Indian invention into global industrial deployment. These ventures, taken together, are one answer: a vertically integrated stack from atom-scale research to capital and commercialization.

Contact

Open to industrial partnerships, capital co-architecture, research collaboration, advisory/board work, and editorial/speaking.

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